

Question Number	Answer	Additional guidance	Mark
1(a)(i)	The only correct answer is A bone <i>B is not correct because ligaments do not attach to cartilage</i> <i>C is not correct because ligaments do not attach to a muscle</i> <i>D is not correct because ligaments do not attach to a tendon</i>		(1)

Question Number	Answer	Additional guidance	Mark
1(a)(ii)	A description that includes the following points: - MRI uses {radio waves / magnetic field} (1) - to produce a {2D / 3D / computer} image (of the anterior cruciate ligament) (1)	ignore radio signals / photograph ACCEPT magnetic radiation / (electro)magnetic {waves / radiation} ignore image unqualified ignore high resolution image	(2)

Question Number	Answer	Additional guidance	Mark
1(a)(iii)	A calculation in which: • calculation of the number of males (1) • calculation of number of males with repairs (1)	<u>Example of calculation:</u> $(67\,000\,000 \div 100) \times 44 = 29\,480\,000$ $(29\,480\,000 \times 77) \div 100\,000 = 22699 / 22700$ Accept in standard form 2.27×10^4 Correct answer with no working scores full marks also another way $77/100000 = .00077$ percentage males with op $0.00077 \times 44/100 \times 67000000$ 22699.6 for 1 mark only NO ECF	(2)

Question Number	Answer	Additional guidance	Mark
1(b)(i)	flexor and extensor	Both required for the mark ACCEPT biceps and triceps mix and match flexor and tricep biceps and extensor ignore extendor	(1)

Question Number	Answer	Additional guidance	Mark
1(b)(ii)	A description that includes two of the following points: • (more) contraction of {A / triceps / extensor} muscle and relaxation of {B / biceps / flexor} muscle (1) • would {pull on / lower / move down} the bones (of the lower arm/radius/ulna) (1) • A and B are antagonistic (pairs) (1)	ignore joint ignore {extending / straightening} the arm	(2)

